



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023– 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE-‘B’

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit II / Operators

Course Outcome: CO 2

Topic Learning Outcome: TLO 5

Activity Chosen: Virtual Laboratory

Justification:

- A Virtual Laboratory serves as an interactive space designed for the creation and execution of simulated experiments, functioning as a playground for exploration and experimentation. It is essentially an on screen simulator or calculator, providing learners with a platform to test ideas and observe outcomes. These virtual laboratories enable learners to engage with and manipulate simulated laboratory equipment, mimicking real-world behavior closely.
- **Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- The instructor provided a detailed explanation of the specific topic in the classroom, covering it comprehensively within a 35-minute session.
- Access to the virtual lab can be obtained through the following link: <https://python-iitk.vlabs.ac.in/> (URL for Virtual Lab).
- Navigate to the Computer Science and Engineering section, which offers a broad range of virtual labs.
- Choose the Python Programming Lab from the list.
- From the drop-down menu, select List of Experiments.
- Opt for the "Arithmetic Operations" experiment, and students were tasked with completing a pretest.
- Clear instructions were given for navigating through the simulator steps.
- The operations were demonstrated using the simulation button.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1
CO 2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Students can be able to apply logical thinking and think about problem-solving approaches.	Students will be able to approach the problem by applying the types of operators they learned and solving simple and complex engineering problems.	Students will be able to interpret the complex engineering problems using the precedence of operators.	The problem solving skill earned through this activity helps the students in solving various problems in information and communication engineering.

- **Images / Screenshot of the practice:**

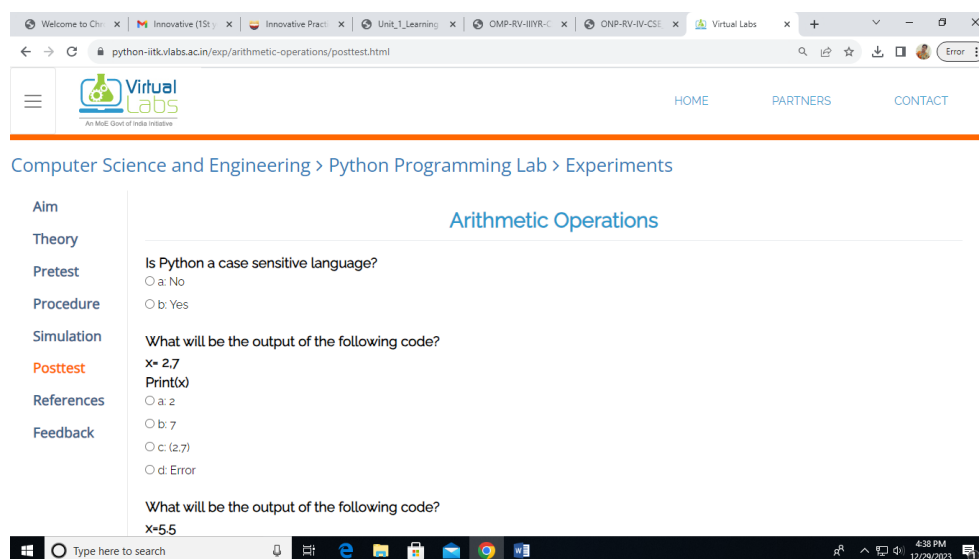


Fig 1: Demonstration on Posttest in operators

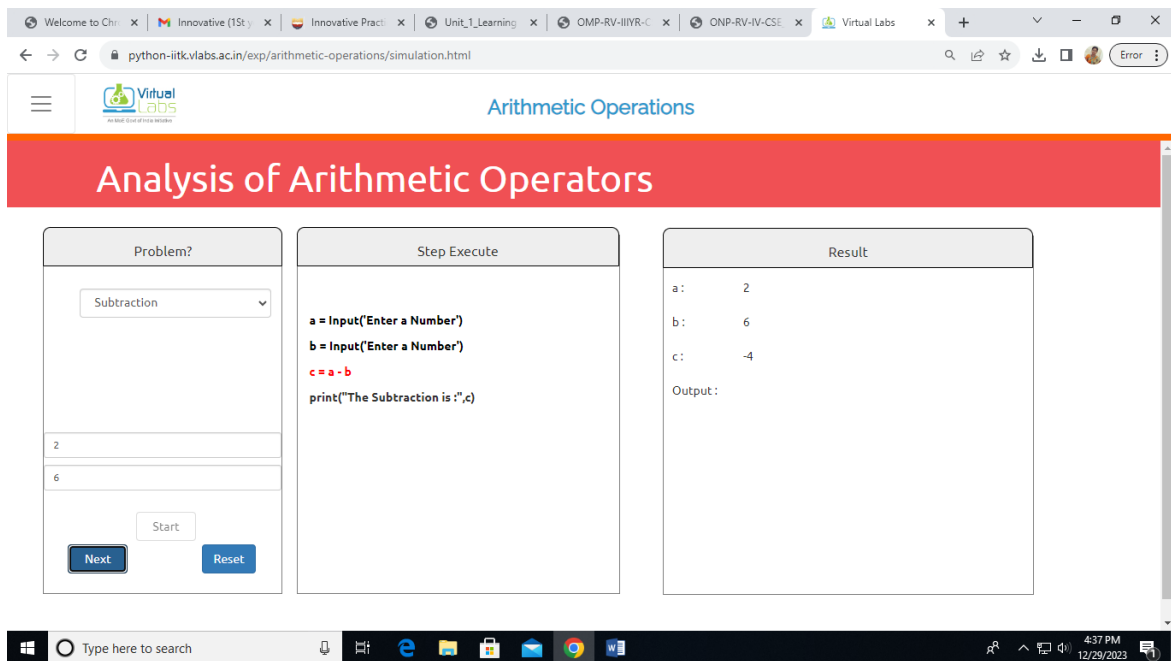


Fig:2 Demonstration on analysis of arithmetic operators

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The practice received positive feedback from students, facilitating their understanding of the fundamentals of Arithmetic Operations in the Python Programming Language through an interactive simulator.
- Students found the pretest and posttest to be valuable tools for reinforcing and recalling key concepts.

❖ **Benefit of the practice:**

- The activity contributed to a clear comprehension of the concept, as expressed by the students.
- Active participation was observed among most students, with a sense of enjoyment, particularly when answering correctly in the pretest.

❖ **Challenges faced in implementation:**

- Some students were less engaged in the demonstration activity.
- The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

1. <https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/index.html>
2. <https://www.ritrjpm.ac.in/images/computer-science/Virtual%20Lab-Routing%20in%20MANET.pdf>
3. https://www.ritrjpm.ac.in/images/computer-science/40.EC8393_virtual%20lab.pdf